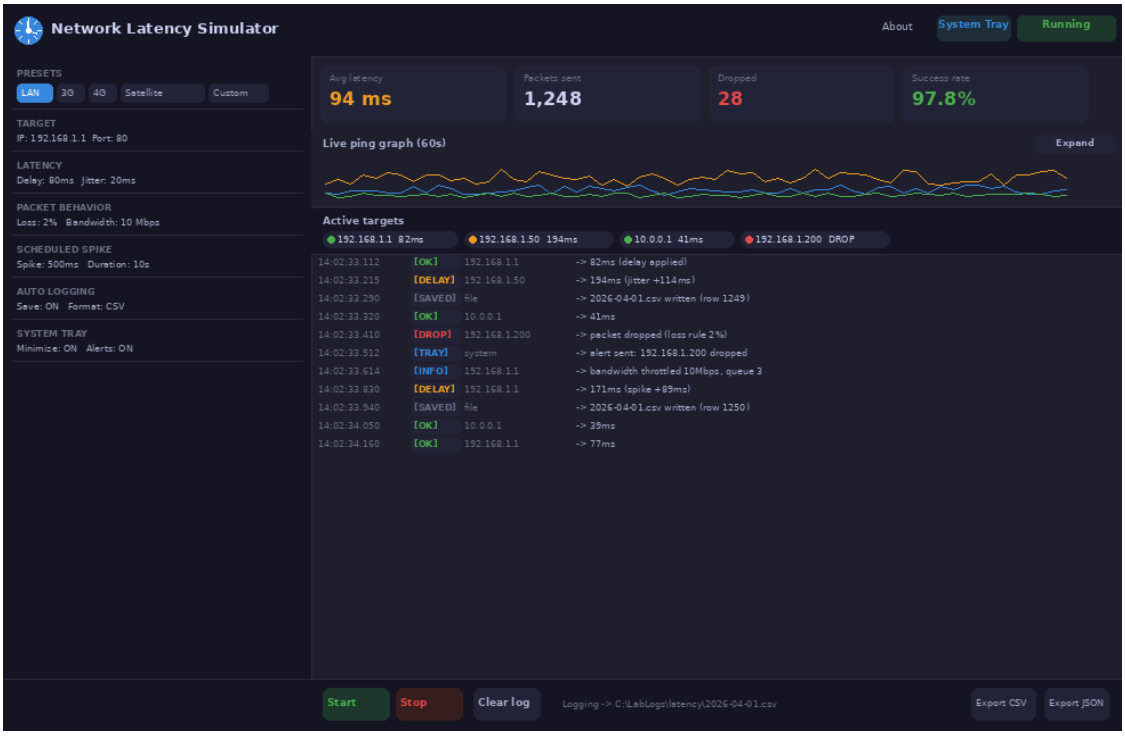
Network Latency Simulator

Complete Project Documentation

Prepared by	Yuvraj Singh Rangi
Designation	Network and Lab Administrator
Institution	PCTE Group of Institutes
Location	Ludhiana, Punjab, India
Year	2026
Version	1.0.0
Technology	Python 3.x, tkinter, matplotlib
Github	@yuvrajsinghrangi



Main application window showing all sections of the Network Latency Simulator

1. Project Overview

The Network Latency Simulator is a desktop application developed using Python and the tkinter graphical user interface framework. It was created by Yuvraj Singh Rangi, Network and Lab Administrator at PCTE Group of Institutes, Ludhiana, Punjab. The tool allows network administrators and IT professionals to simulate real-world network conditions such as high latency, packet loss, jitter, and bandwidth limitations on any target IP address or hostname directly from a computer connected to the campus network.

The primary purpose of this tool is to help the lab administrator test how applications and network-dependent services behave under degraded network conditions, without requiring any external hardware or complex configurations. The software works entirely in software by simulating delays and packet drops before sending probes to a given target. This is extremely useful in an educational lab environment where real network issues need to be studied, demonstrated, and documented.

The application features a modern dark-themed graphical interface divided into a sidebar for controls, a main panel for live data visualization, and a detailed color-coded log panel that records every event in real time. The tool can manage multiple targets simultaneously, draw a live latency graph, export logs in multiple formats, and send desktop notifications when critical events such as packet drops occur.

1.1 Purpose and Goals

The main goal of this project is to give the lab administrator a single easy-to-use tool that replaces the need to use multiple command-line utilities or expensive third-party software. The tool consolidates all necessary functions into one window with a clean graphical interface that is intuitive even for students and junior IT staff who may not be familiar with command-line tools.

A secondary goal is to maintain detailed automatic logs of all simulation sessions so that the administrator can review what happened during a test session at any time. These logs are saved to disk automatically in the format of choice and can be exported for inclusion in reports or presentations at the institution.

2. System Requirements

The application is designed to run on Windows operating systems, which are the predominant platform used across the computer labs at PCTE Group of Institutes. The underlying Python code is cross-platform and can also run on Linux and macOS with minimal changes.

Operating System	Windows 7, Windows 10, Windows 11 or Linux and macOS
Python Version	Python 3.8 or higher
Required Libraries	tkinter, threading, and socket are all built into Python
Optional Libraries	matplotlib for the live graph, pillow for the icon, pystray for system tray
Network Access	Active network adapter and administrator access for ICMP ping
RAM	Minimum 256 MB free RAM for comfortable operation
Storage	Less than 10 MB for the application and all dependencies
Display	Minimum 1024 by 600 screen resolution

To install all required optional libraries, open a command prompt and run the following command.

```
pip install matplotlib pillow pystray
```

3. Application Interface

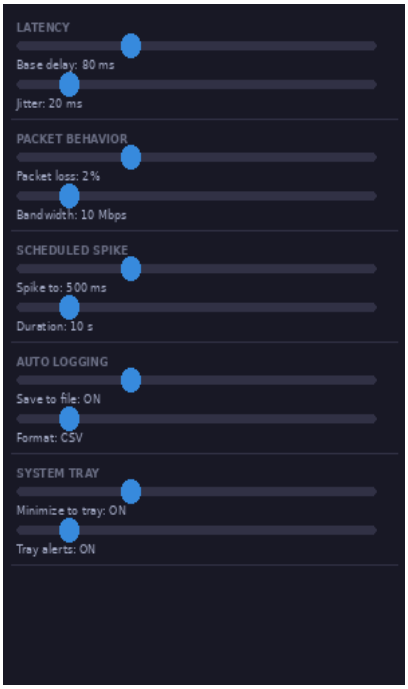
The application window is divided into four distinct sections. The top bar shows the application name, status indicator, and system tray indicator. The left sidebar contains all the simulation controls organized in clearly labeled groups. The main panel on the right shows live statistics, the ping graph, the active targets list, and the live log. The bottom bar provides the start, stop, clear, and export controls.

3.1 Top Bar

The top bar is always visible and shows the application icon on the left followed by the application name. On the right side it shows the About button, a System Tray indicator pill, and a Running or Stopped status pill. The Running pill is displayed in green when the simulation is active and changes to red when stopped. The About button opens a dialog with the author information and version details.

3.2 Sidebar Controls

The sidebar is the control center of the application. It contains all the settings that the administrator can adjust before or during a simulation session. The sidebar can be scrolled vertically when the window is too small to show all controls at once. Controls are grouped into labeled sections for Presets, Target, Latency, Packet Behavior, Scheduled Spike, Auto Logging, and System Tray.



Sidebar panel showing all control groups with sliders, toggles, and input fields

3.3 Main Panel and Statistics

The main panel occupies the right two-thirds of the window. At the top it shows four live statistics cards displaying the average latency, total packets sent, number of dropped packets, and the success rate percentage. These numbers update in real time throughout the simulation. Below the statistics is the live ping graph followed by the active targets bar and the log panel.

Expanded ping graph view showing live latency lines for three simultaneous targets

4. Features in Detail

4.1 Preset Configurations

The preset buttons let the administrator quickly load predefined simulation values. When a preset is selected all the delay, jitter, loss, and bandwidth sliders update automatically to match that network type. The LAN preset simulates a fast local area network. The 3G preset simulates a slow mobile network with 150 milliseconds of base delay and 2 percent loss. The Satellite preset simulates a very high latency connection with 600 milliseconds of base delay.

4.2 Latency and Jitter Simulation

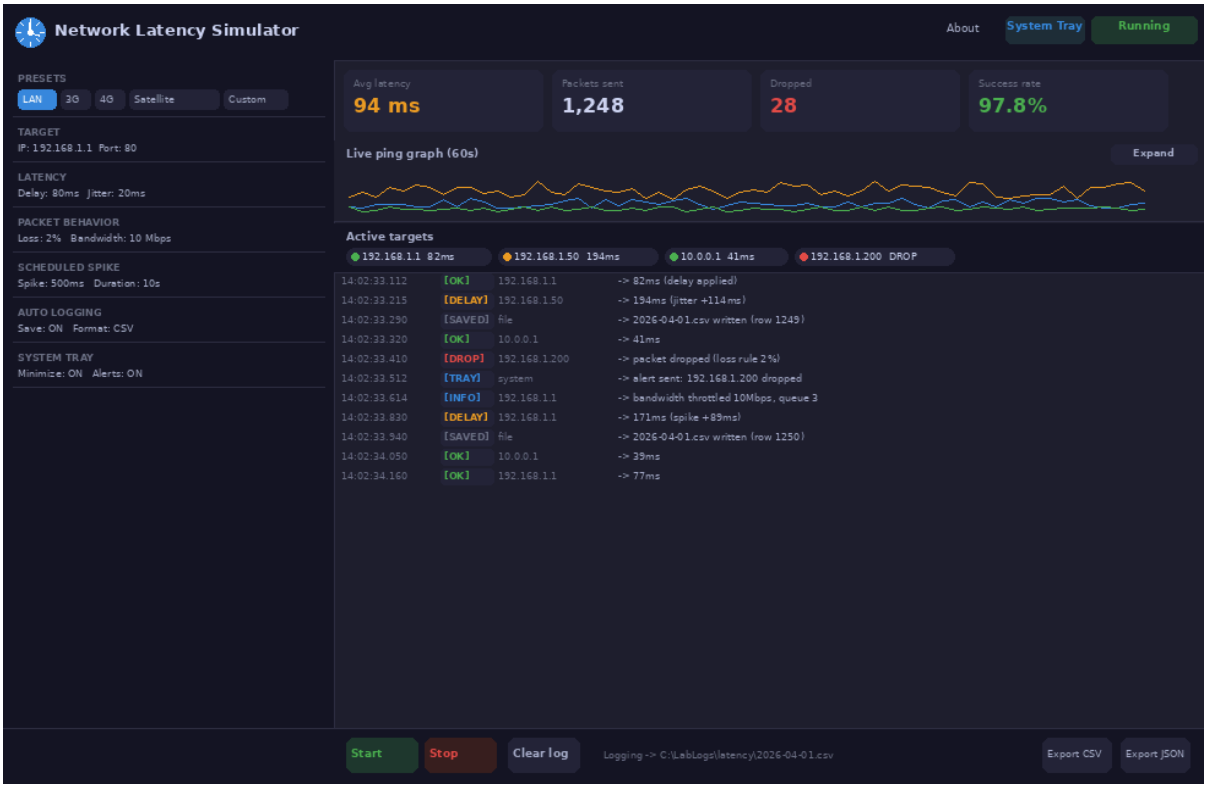
The base delay slider sets the minimum artificial delay added to every packet, configurable from 0 to 2000 milliseconds. The jitter slider adds a random variation on top of the base delay. For example, if the base delay is 80 milliseconds and jitter is 20 milliseconds, each packet experiences a random delay between 60 and 100 milliseconds. This accurately mimics real-world networks where latency is never perfectly consistent.

4.3 Packet Loss and Bandwidth Throttle

The packet loss slider sets the percentage of packets randomly dropped during the simulation. When a packet is dropped the log shows a DROP entry in red with an explanation of which rule triggered it. The bandwidth throttle slider limits the simulated throughput in megabits per second and generates an INFO log entry when the queue depth increases.

4.4 Multiple Target Management

The administrator can add multiple IP addresses or hostnames to be tested simultaneously. Each target runs in its own background thread so all targets are probed independently. Each target is shown as a colored chip in the Active Targets bar. A green dot means the connection is healthy, amber means high latency, and red means packets are being dropped. Each target also gets its own colored line in the live ping graph.



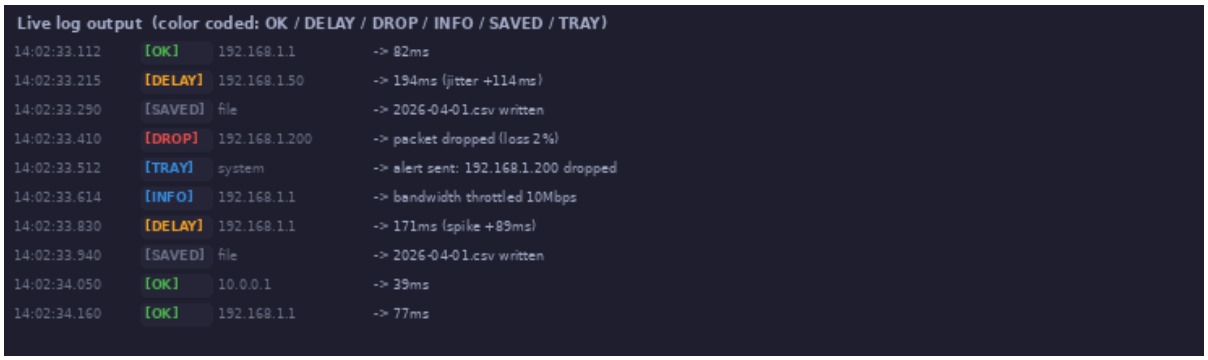
Full main window showing active targets bar with four targets, compact graph, and live log panel

4.5 Scheduled Latency Spike

The scheduled spike feature lets the administrator trigger a sudden latency spike at any point during a running simulation. The Spike To slider sets how high the latency will jump and the Duration slider sets how many seconds the spike will last. Clicking Fire Spike Now activates the spike immediately. After the duration expires the simulation automatically returns to normal base delay and jitter settings.

4.6 Live Log Panel

The live log panel shows a real-time record of every simulation event. Each log row contains a timestamp with millisecond precision, a colored tag showing the event type, the target IP address, and a descriptive message. The log supports up to 500 rows before the oldest entries are automatically removed to keep memory usage low.



Log panel displaying color-coded event rows with OK, DELAY, DROP, TRAY, INFO, and SAVED tags

Feature	Status	Description
OK	Green	Packet sent and response received within normal parameters.
DELAY	Amber	Response received but round-trip time was elevated due to delay or jitter.
DROP	Red	Packet dropped by the loss simulation rule and no response recorded.
INFO	Blue	Informational event such as bandwidth throttling or a spike activation.
SAVED	Gray	A log entry was successfully written to the auto-save file on disk.
TRAY	Blue	A desktop notification was sent to the system tray about a drop event.

4.7 Auto Logging and Export

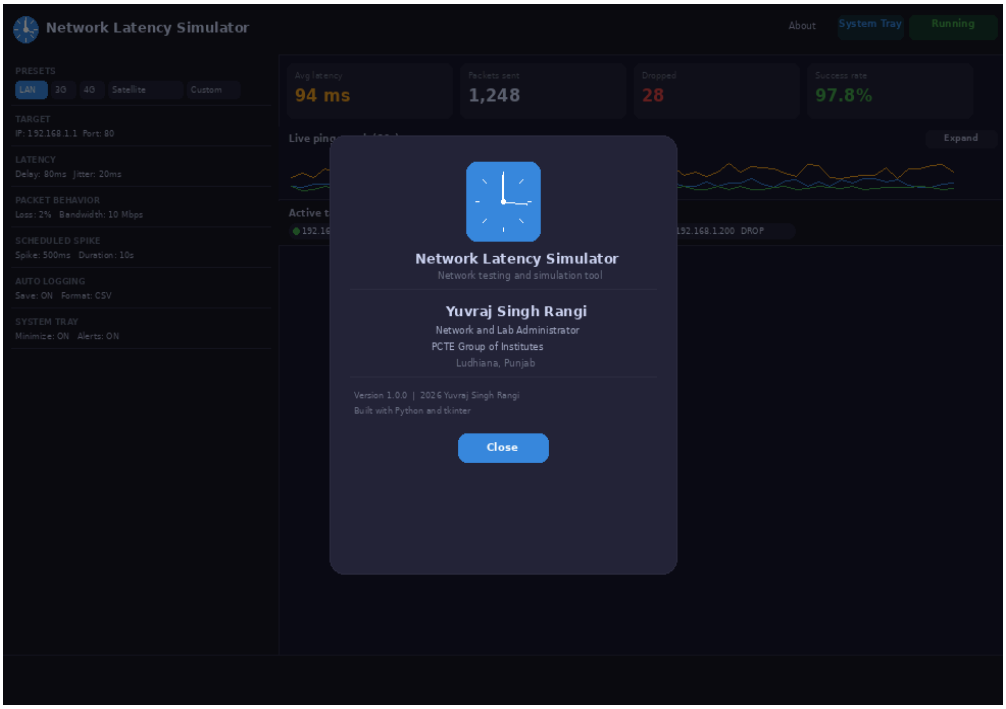
When the Save Log to File toggle is enabled the application automatically writes every packet event to a date-stamped file on disk. The administrator can choose CSV format for Excel compatibility, JSON format for programmatic analysis, or TXT format for a human-readable record. The Export buttons in the bottom bar allow saving the on-screen log to any location at any time.

4.8 System Tray and Background Operation

The Minimize to Tray toggle makes the application hide to the system tray notification area instead of closing when dismissed. The Start with Windows toggle adds the application to the Windows startup registry. The Tray Alerts on Drop toggle sends a desktop balloon notification each time a DROP event occurs, allowing the administrator to be alerted in the background while working on other tasks.

4.9 About Dialog

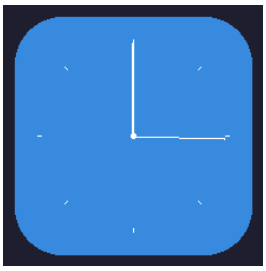
The About dialog is opened by clicking the About button in the top bar. It displays the application icon, application name, author name, institutional affiliation, version number, and copyright year. The dialog confirms the official authorship of the application.



About dialog confirming authorship by Yuvraj Singh Rangi at PCTE Group of Institutes

5. Application Icon

The application uses a custom designed icon showing a clock and radar symbol on a solid blue rounded square background. The icon is embedded in the window title bar, the taskbar, the system tray, and the compiled Windows executable file. The ICO file contains six size variants so Windows can display it clearly at any zoom level or display density.



Custom application icon at 256 by 256 pixels showing the clock and radar design on a blue background

16 x 16 px	Used in window title bars and small taskbar displays
32 x 32 px	Standard taskbar icon size on Windows 10 and earlier
48 x 48 px	Used in folder icon views and medium display modes
64 x 64 px	Used in the application About dialog on screen
128 x 128 px	Used in high DPI taskbar and shortcut displays
256 x 256 px	Used in Windows Explorer and for high resolution printing

6. How to Run and Build

6.1 Running with Python

Open a command prompt in the folder containing latency_simulator.py and install the optional libraries using the following command.

```
pip install matplotlib pillow pystray
```

Then start the application using the following command.

```
python latency_simulator.py
```

On Windows, run the command prompt as Administrator when using ICMP ping mode so the application has permission to create raw network sockets. If administrator access is not available, use the TCP protocol option in the sidebar instead.

6.2 Building the Windows Executable

Place both latency_simulator.py and latency_simulator.ico in the same folder, then run the following PyInstaller command.

```
pyinstaller --onefile --windowed --icon=latency_simulator.ico --name=LatencySimulator
latency_simulator.py
```

The compiled LatencySimulator.exe will appear in the dist folder and can be copied to any lab computer without requiring Python to be installed.

7. Complete Feature Reference

The following table lists every feature available in the application with a description of what each one does and its current implementation status.

Feature	Status	Description
Preset configurations	Included	Five built-in presets for LAN, 3G, 4G, Satellite, and Custom for quick setup.
Base delay slider	Included	Sets the artificial base latency from 0 to 2000 milliseconds added to every packet.
Jitter slider	Included	Adds random variation above and below the base delay to simulate real-world inconsistency.
Packet loss slider	Included	Randomly drops a configurable percentage of packets to simulate network instability.
Bandwidth throttle	Included	Simulates a maximum throughput limit in megabits per second.
Protocol selector	Included	Choose between ICMP ping, TCP, and UDP probe methods for each target.
Multiple targets	Included	Add and remove multiple target IP addresses running in parallel background threads.
Target health chips	Included	Color-coded chips show real-time health status for each active target.
Live ping graph	Included	Matplotlib-powered graph showing 60 seconds of latency history per target.
Graph expand/collapse	Included	Toggle between a compact sparkline view and a fully readable expanded graph view.
Scheduled spike	Included	Fire an instant latency spike to a defined value for a set number of seconds.
Color-coded log	Included	Real-time log with OK, DELAY, DROP, INFO, SAVED, and TRAY event tags.
Auto file logging	Included	Automatically writes log entries to a date-stamped CSV, JSON, or TXT file on disk.
Export log	Included	Save the on-screen log to a user-chosen file at any time.
System tray minimize	Included	Minimize the window to the system tray notification area instead of closing.
Desktop alerts	Included	Send a balloon notification to the system tray when a DROP event is detected.
About dialog	Included	Displays application name, author, institution, version, and build information.
Custom app icon	Included	Professional ICO file with six size variants embedded in the executable.

Dark theme UI	Included	Full dark color scheme for comfortable use in dimly lit lab environments.
Windows .exe build	Included	Compile to a standalone executable using PyInstaller for distribution to lab PCs.

8. Credits and Closing Notes

This application was designed, developed, and documented entirely by Yuvraj Singh Rangi in 2026. It was built to serve the practical needs of the computer lab and network infrastructure at PCTE Group of Institutes, Ludhiana, Punjab, India. The project represents a combination of network engineering knowledge, Python programming skills, and user interface design.

The application is intended for internal use within the institution and may be freely modified or extended by the author to add new features as the needs of the lab evolve. Future enhancements may include support for UDP-based packet generation, a built-in network topology mapper, remote monitoring capabilities, and integration with the existing campus network monitoring dashboard.

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Institution	PCTE Group of Institutes
City	Ludhiana, Punjab, India
Year	2026
Version	1.0.0
Stack	Python 3.x, tkinter, matplotlib, Pillow, pystray
Build Tool	PyInstaller
License	Internal Use Only